

Ayush Verma

Software Development Engineer

Portfolio

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LinkedIn

GitHub

LeetCode

Work Experience

Purple Bean Agro

June 2025 – July 2025

Full Stack Developer Intern

- Developed modular REST APIs for authentication, product, order, and payment workflows using Node.js and Express.js, optimizing query performance through MongoDB indexing and aggregation pipelines.
- Built scalable backend services following a modular architecture, implementing JWT authentication, role-based access control (RBAC), reusable middleware, and efficient database interactions to improve maintainability and security.
- Integrated the Razorpay payment gateway with secure checkout workflows, implementing payment verification, robust API validation, and comprehensive error handling to deliver reliable and secure transaction processing.

Projects

LedgerTrace – Event-Sourced Double-Entry Ledger | Node.js, Express, PostgreSQL, React

[Project Link]

- Designed an **event-sourced**, append-only double-entry ledger where account balances are always derived by replaying the event log, never stored as mutable state – eliminating an entire class of balance-corruption bugs.
- Implemented atomic fund transfers, reversal chains, and point-in-time balance replay, with an independent **reconciliation engine** that detects drift between cached and replayed balances.
- Proved correctness under **concurrency** with a dedicated stress-test suite firing simultaneous transfers against a live server to confirm zero double-spend; deployed on Render (backend) and Vercel (frontend).

Outbox – Email Job Scheduler | Next.js, Express, TypeScript, BullMQ, Redis, PostgreSQL

[Project Link]

- Built a **BullMQ + Redis** delayed-job scheduling engine with a dedicated worker process, verified **restart-persistent** by killing and restarting a live OS process mid-delay with zero job loss.
- Engineered three-layer **idempotency** (job-ID deduplication, database status guards, and a crash-ambiguity state machine) to guarantee at-most-once delivery under partial failure.
- Implemented atomic **rate limiting** via a Redis Lua script enforcing global and per-sender hourly send caps with jittered redelay, validated by 24 infrastructure-level verification assertions in place of mocked unit tests.

Attendy – Real-Time Face Recognition Attendance System | FastAPI, PostgreSQL, pgvector

[Project Link]

- Rebuilt a legacy face-recognition system on **FastAPI** with async SQLAlchemy and **PostgreSQL + pgvector** cosine-similarity search over ArcFace embeddings, eliminating recognizer drift by unifying enrollment and live-recognition capture logic.
- Engineered dual real-time **WebSocket** channels for live recognition streaming and cross-dashboard broadcast, with a per-connection state machine enforcing temporal smoothing (5-of-8 frame consensus) and motion-based liveness detection against spoofing.
- Shipped with Alembic migrations, JWT access/refresh auth, and a **pytest/Vitest** suite validated through **GitHub Actions CI** – documented with an explicit known-limitations section covering isolation and scaling tradeoffs.

Volunteering

- Volunteer, **Google Developer Groups (GDG) On Campus SRM AP**; assisted in organizing technical events, workshops, and hackathons for the student developer community.
- Collaborated with student volunteers to organize workshops, hackathons, and technical events for the developer community.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C++

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, WebSockets

Databases: PostgreSQL, MongoDB, Redis

System Design: Event Sourcing, Idempotency, Distributed Systems, Concurrency Control, Rate Limiting

Developer Tools: Git, GitHub, Docker, Postman, GitHub Actions

Certifications

SAP Certified Generative AI Developer – SAP

[Credential]

MongoDB Associate Developer – MongoDB

[Credential]

Education

SRM University, Andhra Pradesh

2023 – 2027

B.Tech in Computer Science and Engineering

CGPA: 8.6/10.0

Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Achievements and Honors

- Finalist at **SDG Hackathon, HackSRM 7.0**, and **MSC 2.0 Hackathon**.
- Solved **250+** **Data Structures and Algorithms** problems on **LeetCode**, strengthening problem-solving skills.